

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE: Artificial Intelligence (CSA17)

CASE STUDY ANSWERS

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, and logical reasoning for valid conclusions and decision making.

CASE STUDY 1 – Smart Medical Diagnosis System

(a) Propositional Logic Representation

Let: F = Patient has Fever; C = Patient has Cough; B = Patient has Breathlessness; Flu = Possible Flu; M = Possible Measles; P = Possible Pneumonia.

Rules:

1. $F \wedge C \rightarrow \text{Flu}$
2. $F \wedge R \rightarrow M$
3. $C \wedge B \rightarrow P$

Facts about Patient A:

F = True

C = True

B = True

(b) Forward Chaining

Forward chaining starts with known facts and applies rules whose conditions are satisfied.

Step 1: Known facts = F, C, B

Step 2: Rule 1: $F \wedge C \rightarrow \text{Flu}$

Fever and Cough are true.

Therefore Flu = True.

Step 3: Rule 3: $C \wedge B \rightarrow P$

Cough and Breathlessness are true.

Therefore P = True.

Step 4: Rule 2: $F \wedge R \rightarrow M$

Rash is not given.

Therefore Measles cannot be derived.

Final: Possible Flu and Possible Pneumonia.

(c) Backward Chaining for Pneumonia

Goal: Pneumonia(A)

Rule 3:

$\text{Cough}(A) \wedge \text{Breathlessness}(A) \rightarrow \text{Pneumonia}(A)$

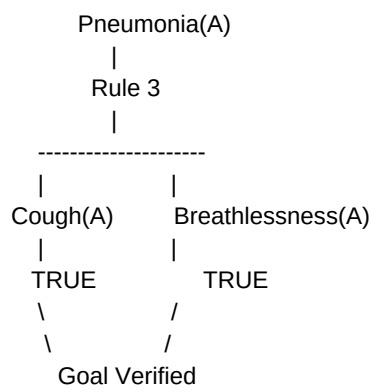
Sub-goals:

$\text{Cough}(A) = \text{TRUE}$

$\text{Breathlessness}(A) = \text{TRUE}$

Both are facts, so:
Pneumonia(A) = TRUE

Goal Reduction Tree:



CASE STUDY 2 – Autonomous Traffic Management Agent

(a) Unification of Rule 1

Rule 1:

$\forall x [\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)]$

Facts:

$\text{Vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

Match $\text{Vehicle}(x)$ with $\text{Vehicle}(\text{Ambulance})$:

$x = \text{Ambulance}$

Match $\text{EmergencyType}(x)$ with $\text{EmergencyType}(\text{Ambulance})$:

$x = \text{Ambulance}$

MGU / substitution:

$\theta = \{x/\text{Ambulance}\}$

Therefore:

$\text{ClearPath}(\text{Ambulance})$

(b) Resolution Proof of Proceed(CarA)

Relevant facts:

$\text{Vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

$\text{Vehicle}(\text{CarA})$

$\text{Behind}(\text{CarA}, \text{Ambulance})$

CNF clauses:

1. $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$
2. $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$
3. $\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$
4. $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x, y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Resolution:

$\text{Vehicle}(\text{Ambulance}) + \text{EmergencyType}(\text{Ambulance})$
 $\rightarrow \text{ClearPath}(\text{Ambulance})$

$\text{ClearPath}(\text{Ambulance}) + \text{Rule 2}$
 $\rightarrow \text{GreenSignal}(\text{Ambulance})$

$\text{Vehicle}(\text{CarA}) + \text{Behind}(\text{CarA}, \text{Ambulance})$
 $+ \text{GreenSignal}(\text{Ambulance}) + \text{Rule 3}$
 $\rightarrow \text{Proceed}(\text{CarA})$

Therefore:

$\text{Proceed}(\text{CarA})$ is TRUE.

(c) Two Simultaneous Emergency Vehicles

The universal variable in Rule 1 allows the rule to apply to multiple vehicles. However, the knowledge base needs facts identifying each emergency vehicle and its position. For example, $\text{Vehicle}(\text{FireTruck})$ and $\text{EmergencyType}(\text{FireTruck})$ would allow $\text{ClearPath}(\text{FireTruck})$ and $\text{GreenSignal}(\text{FireTruck})$ to be derived. Additional conflict-resolution rules are needed if two emergency vehicles require the same path or signal.

CASE STUDY 3 – Agricultural Expert Reasoning System

(a) CNF Conversion

P1:

$\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

$= \neg \text{SoilDry} \vee \text{IrrigationNeeded}$

P2:

$\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

$= \neg(\text{IrrigationNeeded} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$

$= \neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

P3:

$\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

$= \neg(\neg \text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$

$= \text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

Facts:

SoilDry

CropWheat

(b) Resolution Refutation for ApplyDripMethod

Goal: ApplyDripMethod

Negated goal: $\neg \text{ApplyDripMethod}$

P1: $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

Fact: SoilDry

Resolution $\rightarrow \text{IrrigationNeeded}$

P2: $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

Facts: $\text{IrrigationNeeded}, \text{CropWheat}$

Resolution $\rightarrow \text{ApplyDripMethod}$

Now resolve:

$\text{ApplyDripMethod} + \neg \text{ApplyDripMethod}$

$\rightarrow \blacksquare$ (empty clause)

Therefore ApplyDripMethod is proved.

Resolution tree:

SoilDry

|

| P1

↓

IrrigationNeeded

|

| P2 + CropWheat

↓

ApplyDripMethod

|

| + $\neg \text{ApplyDripMethod}$

↓

$\blacksquare$

(c) Remove SoilDry

Without SoilDry :

P1 cannot derive IrrigationNeeded .

Therefore:

IrrigationNeeded cannot be derived.
ApplyDripMethod cannot be derived.

P3 requires:

$\neg \text{ApplyDripMethod} \wedge \text{CropWheat}$

Although CropWheat is known, $\neg \text{ApplyDripMethod}$ is not known.

Therefore CropAtRisk does NOT logically follow.

Conclusion:

IrrigationNeeded: Not derivable

ApplyDripMethod: Not derivable

CropAtRisk: Not derivable

CASE STUDY 4 – Wumpus World Knowledge Agent

(a) *Belief State and Modus Ponens*

Given percepts:

Stench[1,2]

Breeze[1,1]

Glitter[2,2]

Rule 1:

Stench[1,2] $\rightarrow$ WumpusAdjacent[1,2]

Therefore WumpusAdjacent[1,2].

Rule 2:

Breeze[1,1] $\rightarrow$ PitAdjacent[1,1]

Therefore PitAdjacent[1,1].

Rule 3:

Glitter[x,y] $\rightarrow$ Gold[x,y]

Therefore Gold[2,2].

Rule 4:

$\neg$ Wumpus[x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

No negative Wumpus/Pit facts are given, so Safe cannot be derived.

Belief state:

{Stench[1,2], Breeze[1,1], Glitter[2,2],

WumpusAdjacent[1,2], PitAdjacent[1,1], Gold[2,2]}

(b) *Forward Chaining – Possible Locations*

Stench[1,2]

↓

Wumpus adjacent to [1,2]

↓

Possible Wumpus:

[1,1], [1,3], [2,2]

Breeze[1,1]

↓

Pit adjacent to [1,1]

↓

Possible Pit:

[1,2], [2,1]

Glitter[2,2]

↓

Gold[2,2]

(c) *Safety of Path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2]*

Cell [1,1]:

Breeze[1,1] indicates that a pit may be adjacent to [1,1].

The KB does not prove [1,1] is safe.

Cell [1,2]:

Possible Pit locations include [1,2].

Therefore [1,2] may contain a pit.

Stench[1,2] also indicates a nearby Wumpus.

Cell [2,2]:

Glitter proves Gold[2,2], but the KB does not prove:

$\neg$ Wumpus[2,2] and $\neg$ Pit[2,2].

Therefore $[2,2]$ cannot be proven safe.

Final conclusion:

The path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is NOT guaranteed to be safe.

The agent should not assume the path is safe without additional knowledge.

End of Assessment Answers